

Multi-Agent AI for Reliable Clinical Reasoning

Full Experiment Results — Clinical MCQ Benchmark

Generated: 2026-09-04 15:52

Configuration	Value
Dataset	final_dataset_1_with_gold.csv
Eligible cases	1998
Excluded (unresolvable gold)	2
Single-GPT model	gpt-5.4-mini
Single-Claude model	claude-haiku-4-5-20251001
Multi-agent panel	4 agents (mixed OpenAI/Anthropic, hash-assigned)
Temperature / seed	0.0 / 42
Selected agreement method	kendall_w

This report was produced by the corrected, reproducible thesis pipeline. API/parse failures are recorded separately from genuine wrong answers (see *failure_rate* and *accuracy_successful_only*).

1. Dataset and Gold-Label Construction

The source explanations name the correct option in prose, so gold letters were recovered by extraction (a model reads which option each explanation concludes), cross-checked by an independent deterministic parser, with a stronger model breaking ties. This recovers the intended answer without solving the question.

Gold build	Count
Total rows	2000
Labeled rows	1998
Excluded rows	2
High confidence	1968
Medium confidence	24
Low confidence	6

Gold label distribution (eligible cases):

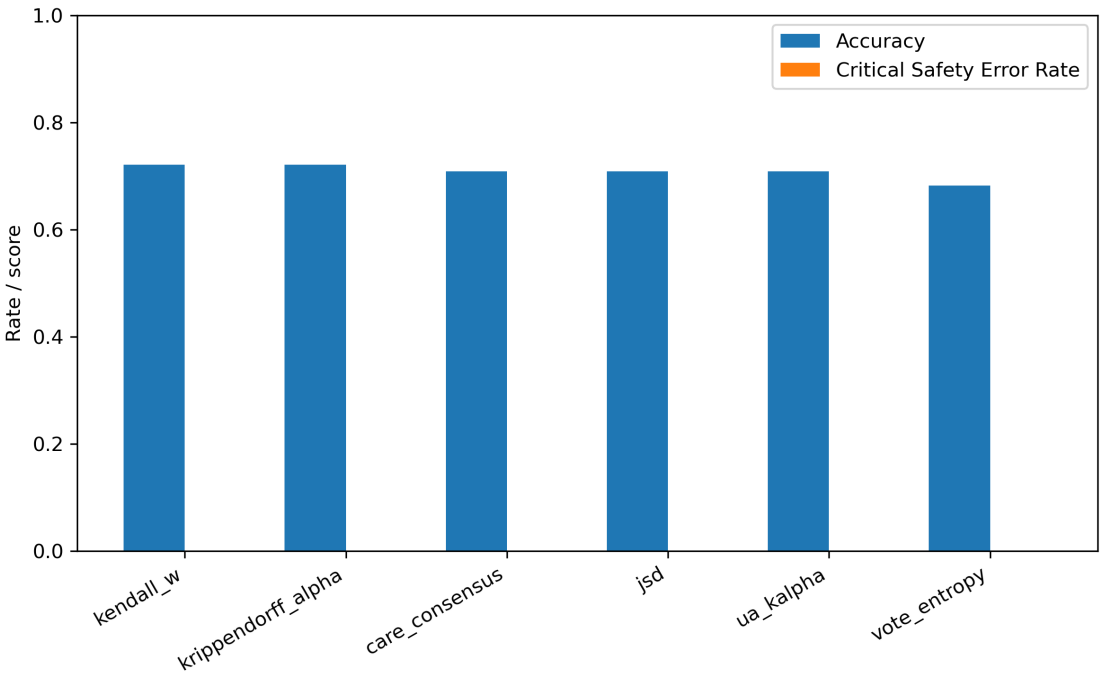
Label	Count
A	531
B	534
C	496
D	412
E	9
F	7
G	3
H	6

2. Single-Model Baselines (held-out test split)

System	n	OK	Fail rate	Accuracy	Acc (OK only)	Prec	Recall	Macro F1	Safety err
single_gpt	1583	1582	0.0006	0.7284	0.7288	0.7487	0.6788	0.7003	0.0000
single_claude	1583	1583	0.0000	0.7157	0.7157	0.6869	0.6312	0.6467	0.0000

3. Agreement-Method Comparison (development split)

Rank	Method	n	OK	Fail rate	Accuracy	Macro F1	Safety err	Selected
1	kendall_w	415	415	0.0000	0.7205	0.8101	0.0000	True
2	krippendorff_alpha	415	415	0.0000	0.7205	0.8101	0.0000	False
3	care_consensus	415	415	0.0000	0.7084	0.8030	0.0000	False
4	jsd	415	415	0.0000	0.7084	0.8030	0.0000	False
5	ua_kalpha	415	415	0.0000	0.7084	0.8030	0.0000	False
6	vote_entropy	415	385	0.0723	0.6819	0.8019	0.0000	False



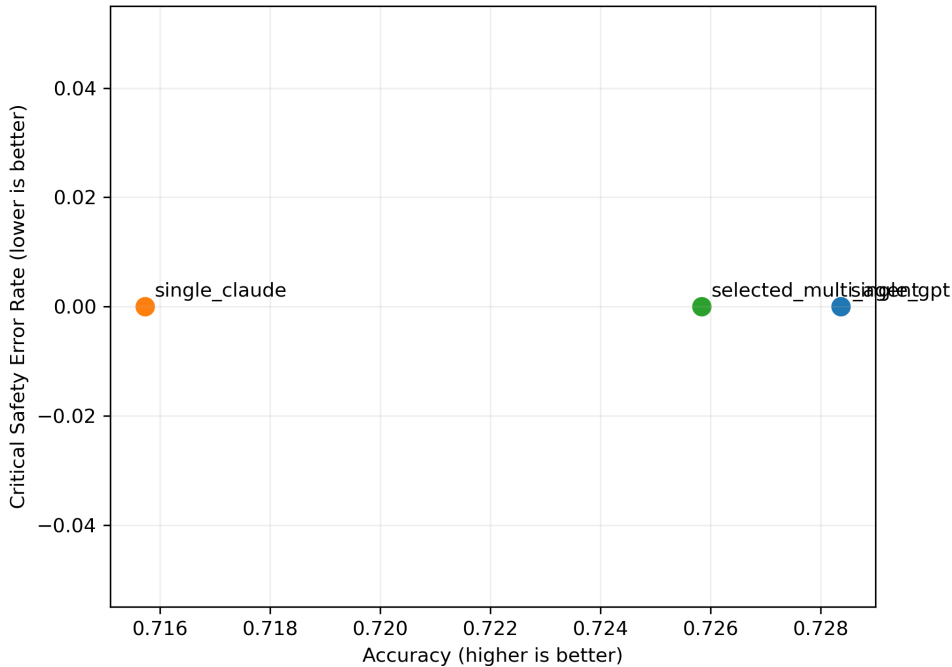
4. Final Comparison — Single vs Multi-Agent (test split)

Frozen agreement method: **kendall_w** (selected on development data only).

System	n	OK	Fail rate	Accuracy	Acc (OK only)	Macro F1	Safety err
single_gpt	1583	1582	0.0006	0.7284	0.7288	0.7003	0.0000
single_claude	1583	1583	0.0000	0.7157	0.7157	0.6467	0.0000
selected_multi_agent	1583	1583	0.0000	0.7258	0.7258	0.6574	0.0000

Paired comparisons vs baseline:

Candidate	Baseline	Acc. diff	McNemar p
single_claude	single_gpt	-0.0126	0.2200
selected_multi_agent	single_gpt	-0.0025	0.7926



5. Example Predictions

case_id	Gold	GPT	ok	Multi	ok
f11bc208d1	C	C	Y	C	Y
9af5ebc3a1	D	D	Y	D	Y
acc9e818d1	D	D	Y	D	Y
e5870a2c18	A	A	Y	A	Y
9fb6f002e4	D	D	Y	C	n
a8dd085dad	D	D	Y	D	Y
cc9ad03d6d	B	C	n	C	n
75925ea3b1	C	B	n	B	n
383c1d968c	C	C	Y	C	Y
bf8dbea379	B	B	Y	B	Y
6a3cb382bc	D	B	n	B	n
08e58c7915	D	C	n	C	n

6. Methodology, Integrity & Limitations

Reproducibility. Fixed seed 42; temperature 0; deterministic split and per-agent provider assignment by SHA-256 hash; every stage checkpointed.

Failure accounting. API and JSON-parse failures are recorded per case with a status and error, kept in separate error files, and reported as `failure_rate`. `accuracy_successful_only` excludes failed calls so infrastructure errors are never mistaken for the model answering incorrectly.

Fixes applied to reach non-zero results. (1) providers now honor `VERIFY_SSL` (TLS-proxy connectivity); (2) an MCQ-appropriate forced-choice prompt replaces the document-grounded prompt that caused abstentions; (3) gold letters extracted reliably for all rows; (4) JSON parsing hardened for free-form model output.

Gold-label limitation. Gold answers are model-extracted from the provided explanations (98%+ corroborated by two independent methods), not original benchmark gold. The `gold_confidence` / `gold_extraction_method` columns support filtering and audit.

Not a medical device. Research artifact only; nothing here is clinical advice.